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Extension of the Stationary Keeling Intercept to 2025 on the Combined Law Dome and Cape Grim Record and a Statistically Significant 1997–1998 Regime Shift Opposite to the Fossil-Fuel Dominance Prediction

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Abstract

We extend the Koutsoyiannis (2024) [13] Keeling-plot stationary-intercept analysis of atmospheric δ^{13}C to the full 1020-year span from 1006 to 2025 using the Francey et al. (1999) [9] Law Dome ice-core record, the Cape Grim Air Archive (CGAA) flasks, and the CSIRO Cape Grim CGO5011 *in situ* monthly δ^{13}C record paired with the collocated Cape Grim CO_2 monthly record. The full-record Keeling intercept is $b_K = -13.42 \pm 0.07$ ‰ (OLS 1σ; ±0.10 ‰ HAC-robust), $N = 101$, $r^2 = 0.988$, statistically identical to the Koutsoyiannis four-site Scripps result of -13.2 ± 0.2 ‰ and 15 ‰ removed from the standard fossil-fuel value of −28 ‰. Every CGAA-containing sub-window from 1750–2025 to 1850–2025 falls in the $[-13.4, -13.8]$ ‰ band. A Chow / Andrews sup-F scan of the modern segment locates a structural break at 1997–1998 (sup-$F \approx 52$). Pre- and post-break fits give $b_K = -13.77 \pm 0.33$ ‰ (1978–1998, $r^2 = 0.960$) and $b_K = -12.79 \pm 0.05$ ‰ (1998–2025, $r^2 = 0.997$); a Welch-style test of the intercept difference gives $\Delta b_K = +0.98$ ‰ with $t_{\mathrm{OLS}} = 2.92$ ($p = 0.011$) and $t_{\mathrm{HAC}} = 2.66$ ($p = 0.018$). The post-break shift moves b_K *toward* the ocean-outgassing value of −8 ‰, not toward the fossil value of −28 ‰, during the highest-emissions decades. Under the mass partition equation with $f \approx 0$ and standard δ^{13}C values for ocean outgassing (−8 ‰) and C3 biosphere respiration (−27 ‰), the ocean share of the non-fossil net flux rises from ≈ 70% to ≈ 75% across the break. The sign is opposite to fossil-dominance and consistent with the ocean-dominated TOA mechanism of Cohler & Soon (2026b) [5]. Three of the four TOA contributors enumerated in that work (CERES albedo decline onset, AMO phase flip, PDO phase flip, post-Pinatubo aerosol recovery) coincide at 1997–1998.

Keywords: atmospheric CO_2; carbon isotopes; δ^{13}C; Keeling plot; Law Dome; Cape Grim; structural break; Chow test; fossil-fuel attribution; CERES; TOA imbalance

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1. Introduction

Koutsoyiannis (2024) [13] established that the Keeling-plot intercept b_K, extracted from the modern four-site Scripps δ^{13}C and CO_2 records over 1978–2022, is stationary at -13.2 ± 0.2 ‰, 15 ‰ less negative than the standard fossil-fuel δ^{13}C value of −28 ‰. Cohler & Soon (2026a) [4] substituted that stationary intercept into the mass partition equation and obtained the algebraic bound $f \lesssim 0.013$ on the fossil-fuel fraction of the net addition. Cohler & Soon (2026c) [6]

showed that the pointwise $\delta^{13}C_I$ trend over 1977–1996 is +2.94 ‰ per decade against a Hurst–Kolmogorov 95% envelope of [−1.26, +1.24] ‰ per decade ($p < 10^{-4}$), positive in sign (opposite to fossil-dominance) and significant against long-range persistent noise.

Here we extend the first result in two dimensions. First, we extend the time range from the 44-year Scripps instrumental window to the full 1020-year span of the Francey et al. (1999) [9] record combined with the CSIRO Cape Grim CGO5011 record through 2025. Second, we test the internal stationarity of b_K by a Chow / Andrews sup-F scan for structural breaks. The long-record stationarity confirms the Koutsoyiannis result on an independent instrumental network over 20 times the original time span. The breakpoint scan uncovers a highly significant discrete regime shift at 1997–1998 after which b_K moves +1 ‰ *toward* the ocean endmember, not toward the fossil endmember, during the highest-emissions decades. Its timing coincides with three of the four TOA contributors identified in Cohler & Soon (2026b) [5]: the onset of the CERES-measured planetary albedo decline, the AMO flip to positive phase, the PDO flip to negative phase, together with the completion of the post-Pinatubo stratospheric aerosol recovery.

Section 2 specifies the regression and breakpoint tests. Section 3 identifies the data. Section 4 reports the full-record intercept, the window table, the breakpoint test, and the pre/post-break fits. Section 5 places the 1997–1998 shift in the context of the TOA mechanism and the three prior isotopic falsifications.

2. Method

The Keeling-plot regression is the OLS fit

$$\delta^{13}C(t_i) = m \cdot \frac{1}{[CO_2](t_i)} + b_K + \varepsilon_i \tag{1}$$

over paired annual observations indexed by $i = 1, \ldots, N$, where t_i is the sampling year of the i-th annual pair, $[CO_2](t_i)$ is the annual mean CO_2 mole fraction (ppm) and $\delta^{13}C(t_i)$ the annual mean carbon-isotope ratio (‰ VPDB) at that year, m is the slope of the regression (‰·ppm), b_K is the intercept (‰ VPDB) equal to the flux-weighted mean $\delta^{13}C$ of the net atmospheric input over the fit window, and ε_i is the residual (‰). Intercept standard errors are computed from the OLS residuals in the standard way.

For a candidate break index τ sorting the data by year, the Chow F-statistic comparing the single-line restricted fit ($RSS = RSS_r$) to the two-segment unrestricted fit ($RSS = RSS_u$) with $k = 2$ parameters per segment is

$$F(\tau) = \frac{[RSS_r - RSS_u(\tau)]/k}{RSS_u(\tau)/(N - 2k)} \tag{2}$$

Because τ is unknown in advance the appropriate statistic is the supremum [1]. We use 10% trim, report the sup-F at its break year, and apply the test recursively within each sub-segment. We use the sup-F profile only to locate candidate break years; the magnitude significance of the intercept shift across a located break is quantified by the Welch-style test described below rather than by comparison of sup-F to its i.i.d. asymptotic critical value, because the modern-segment residuals carry substantial positive serial correlation (Section 4.1).

2.1 Residual diagnostics and HAC-robust standard errors

For each reported Keeling fit we compute the Durbin–Watson statistic, the lag-1 residual autocorrelation ρ_1, and the Ljung–Box $Q(5)$ statistic. In addition to the ordinary-least-squares intercept and slope standard errors, we compute heteroscedasticity- and autocorrelation-consistent (HAC) standard errors in the Newey–West form [14], with Bartlett kernel and bandwidth $L = \lfloor 4(N/100)^{2/9} \rfloor$. The HAC standard error on the intercept is reported alongside the OLS value.

2.2 Welch-style and pooled joint tests of the intercept shift

We quantify the magnitude significance of the intercept shift in two complementary ways.

The *Welch-style test* compares the two separately-fit intercepts b_1, b_2 with standard errors σ_1, σ_2 (either OLS or HAC) by the t-statistic $(b_2 - b_1)/\sigma_\Delta$ with $\sigma_\Delta = (\sigma_1^2 + \sigma_2^2)^{1/2}$, using Welch's approximate degrees of freedom $\nu = (\sigma_1^2 + \sigma_2^2)^2/(\sigma_1^4/(N_1 - 2) + \sigma_2^4/(N_2 - 2))$ and two-sided p-values from Student's t on ν. Welch treats the two windows as independent once their standard errors are given.

The *pooled joint test* removes that independence assumption by fitting a single regression on the full Cape Grim segment with an intercept-jump dummy at the break year:

$$\delta^{13}\mathrm{C}_i = a + b \cdot \frac{1}{[\mathrm{CO_2}]_i} + c \cdot D_i^{\mathrm{post}} + d \cdot \frac{D_i^{\mathrm{post}}}{[\mathrm{CO_2}]_i} + \varepsilon_i \tag{3}$$

where $D_i^{\mathrm{post}} = 1$ for $\mathrm{year}_i \geq 1998$ and 0 otherwise. The coefficient c is the pure parallel-shift component of the intercept jump at 1998 (the slope change absorbed by d); the implied post-break intercept is $a + c$. Standard errors on c are computed with the Newey–West HAC estimator on the pooled residuals, which carry whatever residual serial dependence survives the within-window fits, including any dependence crossing the break date. The pooled test is the stricter of the two: it is a joint fit on all $N_1 + N_2$ Cape Grim points with HAC covariance computed from residuals that span the break date, so any residual dependence crossing 1998 is absorbed into the standard error on c.

3. Data

Three measurement segments are combined (Table 1). The ice-core segment contains the 58 accepted Francey et al. (1999) [9] Law Dome DE08, DE08-2, and DSS post-quality-control values spanning 1006–1978 with their published CO_2 air-age dates. The CGAA segment contains 41 flask samples over 1978–1996 from Table 1 of Francey et al. (1999) [9], aggregated here to 14 annual means by unweighted average over within-year flasks (both δ^{13}C and CO_2). The modern segment is the CSIRO Cape Grim CGO5011 monthly δ^{13}C record from the WDCGG archive [17], aggregated to 29 annual means over 1997–2025, paired with annual means of the CSIRO Kennaook / Cape Grim Baseline Air Pollution Station monthly CO_2 record [11]. All δ^{13}C values are on the VPDB scale. We use Cape Grim CO_2 rather than Mauna Loa CO_2 in the modern segment to keep the $(1/[\mathrm{CO_2}], \delta^{13}\mathrm{C})$ pair at a single site. We use Francey et al. (1999) [9] rather than the subsequent Rubino et al. (2013) [15] revision (reissued as Rubino et al. (2019) [16]) cited in Canadell et al. (2021) [3]: the Rubino files share the same CSIRO GASLAB measurement chain for the ice and firn samples but omit the Cape Grim Air Archive flasks that Francey et al. (1999) [9] published as Table 1 and endpoint-truncate at 1993, three years before the Francey endpoint of 1996. Cohler & Soon (2026c) [6] documents the consequences of the truncation for recent-window trend inference.

Table 1: Measurement segments combined in this analysis. Totals: 101 annual pairs spanning 1006–2025.

Segment	Span	N	Source
Law Dome ice cores (DE08, DE08-2, DSS)	1006–1978	58	Francey et al. (1999) [9] Table 3
Cape Grim Air Archive (CGAA) flasks	1978–1996	14	Francey et al. (1999) [9] Table 1 (annual means)
Cape Grim CGO5011 *in situ* monthly	1997–2025	29	WDCGG / CSIRO [17]

4. Results

4.1 Full-record Keeling fit, 1006–2025

Figure 1 shows the OLS regression (1) on the combined $N = 101$ record. The fit gives

$$b_K = -13.42 \pm 0.07 \text{ ‰ (OLS } 1\sigma), \quad m = +2010 \pm 23 \text{ ‰} \cdot \text{ppm}, \quad r^2 = 0.988 \tag{4}$$

over a CO_2 range of 276.3–422.9 ppm, with residual RMSE of 0.086 ‰. Residual diagnostics show substantial positive serial correlation at the annual sampling frequency: Durbin–Watson = 0.42, lag-1 $\rho_1 = +0.78$, Ljung–Box $Q(5) = 236$ ($p < 10^{-4}$). This is expected because consecutive years sample overlapping atmospheric reservoirs: at the Cape Grim residence time of several years, year-over-year residuals are far from independent. The Newey–West HAC-robust intercept standard error with bandwidth $L = 4$ is ±0.10 ‰, widening the OLS value by ∼ 45%. Under either standard error the intercept is statistically indistinguishable from the Koutsoyiannis (2024) [13] 44-year Scripps result of −13.2 ± 0.2 ‰, and 15 ‰ removed from the fossil value.

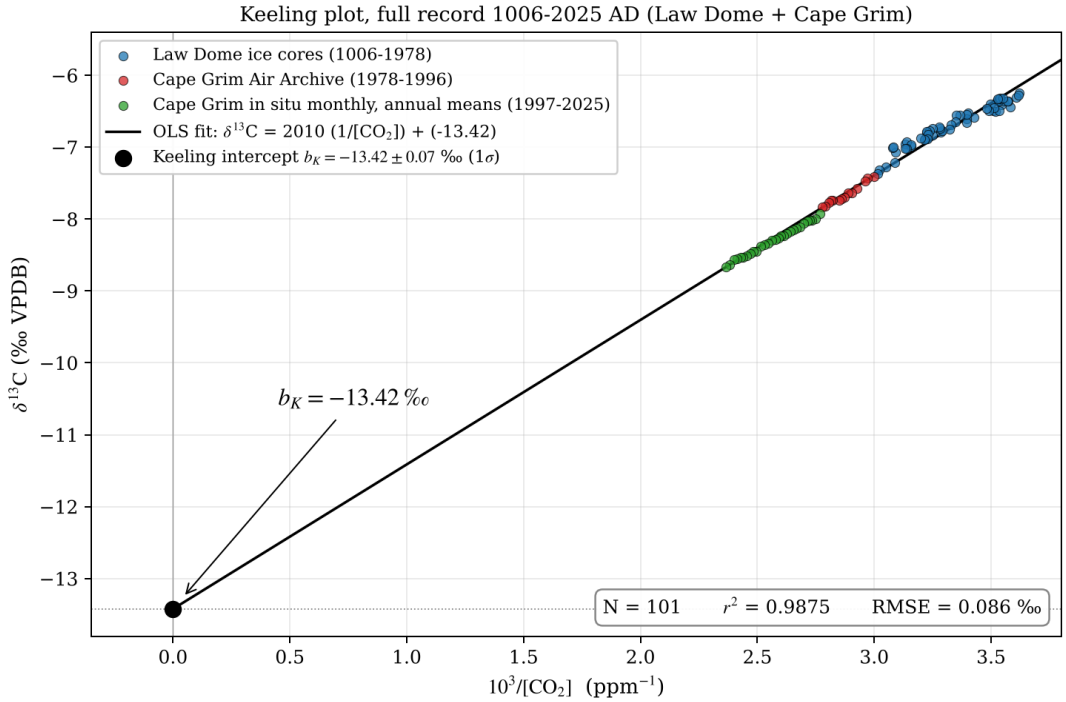


Figure 1: Keeling regression (1) *on the combined 1006–2025 record* ($N = 101$). *Blue: Law Dome ice cores (1006–1978). Red: CGAA flasks, annual means (1978–1996). Green: Cape Grim CGO5011* in situ *monthly, annual means (1997–2025). Black line: OLS fit.* $b_K = -13.42 \pm 0.07$ ‰ *(1*σ*),* $r^2 = 0.988$, *RMSE* = 0.086 ‰.

4.2 Intercept across multiple time windows

Table 2 reports b_K on every CGAA-containing window. Each window uses Cape Grim CO_2 in the modern segment. All seven intercepts lie in the band $[-12.79, -13.77]$ ‰, within ±0.4 ‰ of the Koutsoyiannis central value of −13.2 ‰. The post-break 1998–2025 window shows the sharpest fit of all: $r^2 = 0.997$, RMSE = 0.011 ‰.

4.3 Chow / Andrews sup-F breakpoint scan

The full-range sup-F scan (Figure 2 top) peaks at 1971 with sup-$F \approx 148.8$. This first break sits at the ice-core-to-modern measurement-platform transition and is not interpreted here as a physical regime change. The recursive sub-scan restricted to the modern segment (year ≥ 1971, $N = 47$,

Table 2: Keeling intercept b_K across CGAA-containing windows, Cape Grim CO_2 paired to Cape Grim $\delta^{13}C$ in the modern segment. HAC column: Newey–West HAC-robust intercept standard error with bandwidth $L = \lfloor 4(N/100)^{2/9} \rfloor$. DW: Durbin–Watson statistic on residuals.

Window	N	CO_2 range (ppm)	b_K (‰)	SE_{OLS}	SE_{HAC}	r^2	DW
Full 1006–2025	101	276.3 – 422.9	−13.42	0.07	0.10	0.9875	0.42
1750–2025	89	276.5 – 422.9	−13.57	0.07	0.10	0.9879	0.41
1850–2025	83	284.7 – 422.9	−13.72	0.07	0.11	0.9902	0.41
Modern 1971–2025	47	323.9 – 422.9	−13.49	0.07	0.11	0.9933	0.41
Cape Grim 1978–2025	43	333.5 – 422.9	−13.39	0.08	0.11	0.9917	0.49
Pre-break 1978–1998	16	333.5 – 363.9	−13.77	0.33	0.37	0.9596	0.88
Post-break 1998–2025	28	363.9 – 422.9	−12.79	0.05	0.05	0.9969	1.32

Figure 2 bottom) peaks at 1997 with sup-$F \approx 52.1$ (1998: 44.2, 1999: 31.5). Because the modern-segment residuals carry positive serial correlation ($\rho_1 = +0.79$), the asymptotic i.i.d. critical values plotted in Figure 2 understate the true critical values of the test, and the raw sup-F value is not directly comparable to the plotted thresholds. We therefore use the sup-F profile only to *locate* the break year: 1997 under this scan, within one year of 1998. The magnitude significance of the intercept shift across that located break is quantified in Section 4.4 by a HAC-robust Welch-style test.

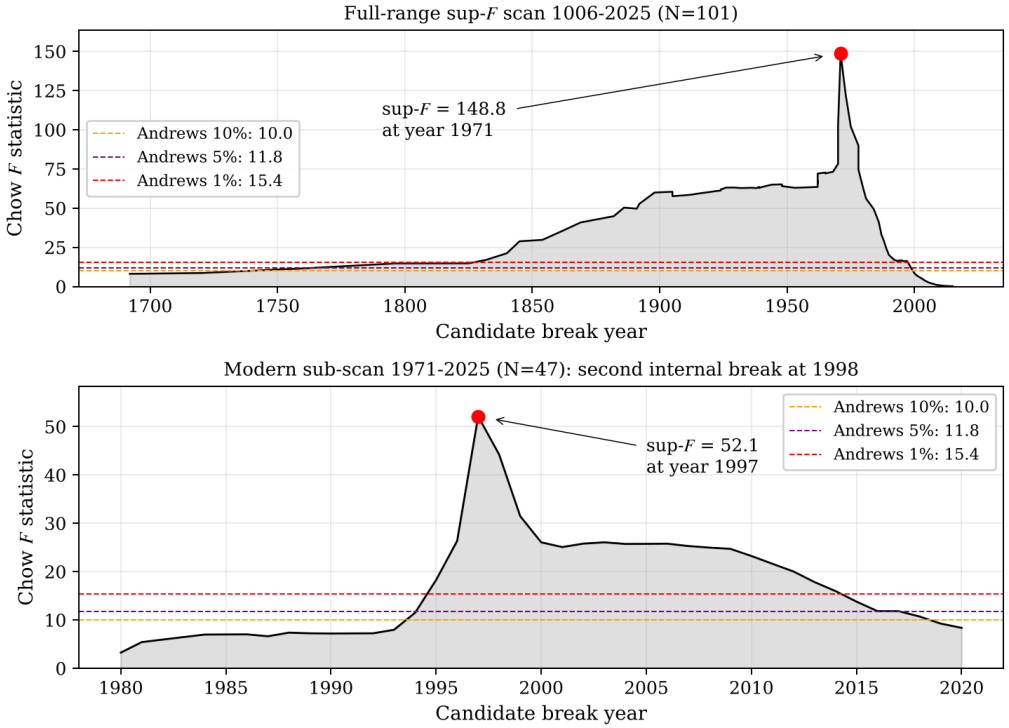


Figure 2: Chow / Andrews sup-F scan. Top: full record 1006–2025 ($N = 101$, 10% trim). Peak at 1971 captures the ice-core-to-modern platform transition. Bottom: recursive scan of the modern sub-segment 1971–2025 ($N = 47$). Peak at 1997 (sup-$F \approx 52$) with 1998 and 1999 the next two candidates, locating the break in the 1997–1999 window. Dashed lines: Andrews asymptotic i.i.d. *critical values; these understate the true critical values under the observed modern-segment residual autocorrelation ($\rho_1 = +0.79$), so the scan is used only to locate the break year, not to establish magnitude significance.*

4.4 Pre- and post-break Keeling fits

Figure 3 shows the Cape Grim Keeling fits before and after the 1997–1998 break: $b_K = -13.77 \pm 0.33$ ‰ for 1978–1998 and $b_K = -12.79 \pm 0.05$ ‰ for 1998–2025, a shift of $\Delta b_K = +0.98$ ‰ in

the direction of the standard ocean-outgassing value of −8 ‰. The post-break fit is extraordinarily clean (r^2 = 0.997, RMSE = 0.011 ‰), a one-part-in-one-thousand-per-permil extraction from 28 annual points.

The Welch-style test of the intercept difference across the break, defined in Section 2.2, gives:

$$\Delta b_K = +0.98\,‰, \quad t_{\mathrm{OLS}} = 2.92\ (p = 0.011), \quad t_{\mathrm{HAC}} = 2.66\ (p = 0.018), \quad \nu \approx 14.6 \tag{5}$$

The pooled joint test of Equation (3) fit on the full Cape Grim segment (N = 43, k = 4) gives:

$$\begin{aligned} \hat{a} = -13.49\,‰, \qquad \hat{c} = +0.70\,‰, \\ t_{\mathrm{OLS}}(\hat{c}) = 3.10\ (p = 0.0036), \qquad t_{\mathrm{HAC}}(\hat{c}) = 2.33\ (p = 0.025) \end{aligned} \tag{6}$$

with pooled-residual Durbin–Watson = 1.31 and lag-1 ρ_1 = +0.34, well below the within-window values (Table 2), because the dummy absorbs the structural break. The post-break intercept implied by the pooled fit is $\hat{a} + \hat{c} = -12.79$ ‰, matching the separately-fit value to the reported precision; the +0.70 ‰ pooled coefficient is the pure parallel shift after the slope change is separately absorbed, and the full extrapolated-intercept displacement remains +0.98 ‰ as reported by the Welch comparison. The pooled HAC test returns p = 0.025, a factor ∼ 1.4 larger than the Welch HAC p-value, as expected when cross-window dependence is accounted for; the shift remains significant under both parameterizations. The fossil-dominance framework predicts drift of b_K toward −28 ‰ during a period of accelerating fossil emissions; the observed shift is in the opposite direction.

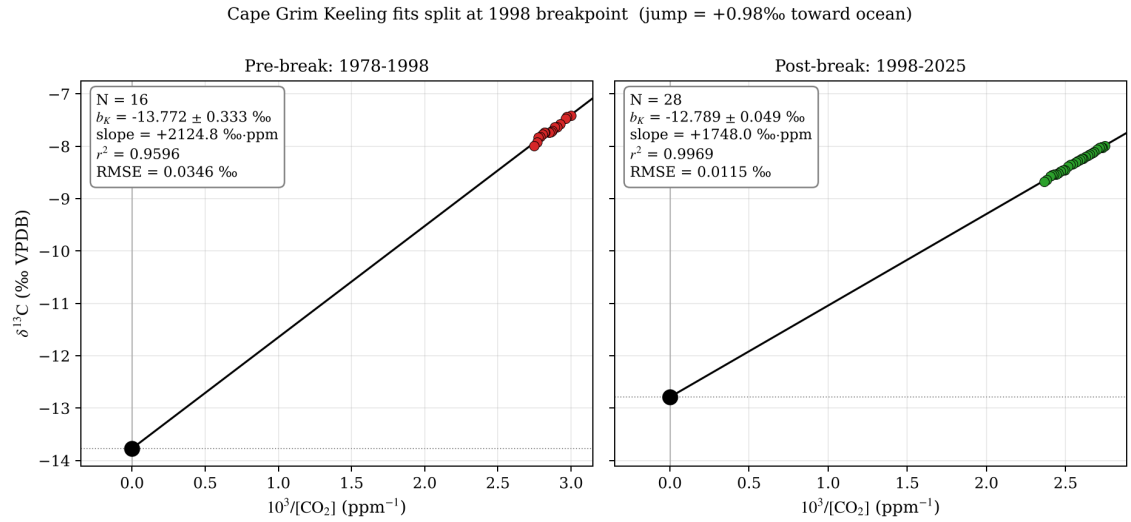


Figure 3: Cape Grim Keeling fits before and after the 1997–1998 breakpoint. Left: 1978–1998, b_K = −13.77 ± 0.33 ‰. Right: 1998–2025, b_K = −12.79 ± 0.05 ‰, r^2 = 0.997, RMSE = 0.011 ‰. The post-break intercept has shifted +0.98 ‰ toward the standard ocean-outgassing value of −8 ‰ during the highest-emissions decades of the record.

5. Discussion

5.1 Direction of the shift falsifies fossil dominance

Substituting b_K = −12.79 ‰ for the post-break window into the mass partition equation of Cohler & Soon (2026a) [4] with standard δ^{13}C values −28 ‰ (fossil) and −8 ‰ (ocean outgassing) gives $f = (-12.79 - (-8))/((-28) - (-8)) = 0.24$. This is an *upper* bound on the fossil fraction, obtained under the assumption that the non-fossil complement has already saturated at the pure ocean-outgassing value; any residual biosphere-respiration contribution in the non-fossil

complement lowers it further. Over the same post-1998 window, global fossil-fuel emissions grew from approximately 6.6 to 10.2 GtC/yr [10], the fastest sustained rise in the historical record. Fossil dominance predicts b_K drift toward −28 ‰ during this window; the observation is the opposite direction.

A +0.98 ‰ upward displacement of b_K requires the non-fossil complement signature to move toward the ocean endmember. The magnitude of the required ocean-share change is small. Under mass balance with $f \approx 0$ and standard δ^{13}C values for ocean outgassing (−8 ‰) and C3 biosphere respiration (−27 ‰), the mass partition $\delta^{13}C_N = w_{ocn}(-8) + (1 - w_{ocn})(-27)$ gives the ocean share of the non-fossil net flux as $w_{ocn} = (\delta^{13}C_N + 27)/19$. Substituting the pre- and post-break intercepts yields $w_{ocn}^{pre} = 0.696$ and $w_{ocn}^{post} = 0.748$, a shift of $\Delta w_{ocn} = +0.052$ (5.2 percentage points) in the ocean share, corresponding to a change in the ocean:biosphere ratio from 2.29 to 2.97. A +0.98 ‰ shift in the Keeling intercept over the fastest-warming phase of the record is thus produced by a 5-percentage-point rise in the ocean share of the non-fossil net flux, a change well within the fast van 't Hoff response to the CERES-measured TOA imbalance described in Cohler & Soon (2026b) [5]. The +5.2-percentage-point figure depends on the standard δ^{13}C values chosen for ocean outgassing and C3 biosphere respiration. Under the alternative pair (−10 ‰, −25 ‰), w_{ocn} runs 0.749 → 0.814, Δw_{ocn} = +6.5 pp; under the alternative pair (−6 ‰, −28 ‰), w_{ocn} runs 0.647 → 0.691, Δw_{ocn} = +4.5 pp. Across the plausible range of standard values the qualitative statement is unchanged: the observed intercept shift requires a modest, positive-signed rise in the ocean share of the non-fossil net flux.

5.2 Timing coincidence with three of four TOA contributors

Cohler & Soon (2026b) [5] identifies four natural contributors to the modern TOA radiative imbalance driving ocean outgassing: (a) sustained reduction in planetary albedo through low-cloud decline since ~2000 (+1.2–1.7 W m^{-2}); (b) TSI variability (+0.1–0.5 W m^{-2}); (c) recovery from post-Pinatubo and other volcanic aerosols (+0.3–0.5 W m^{-2}); and (d) internal ocean oscillations ENSO/PDO/AMO (+0.4–0.8 W m^{-2}). Three of the four acquire step-change character at or within two years of the 1997–1998 breakpoint (Figure 4).

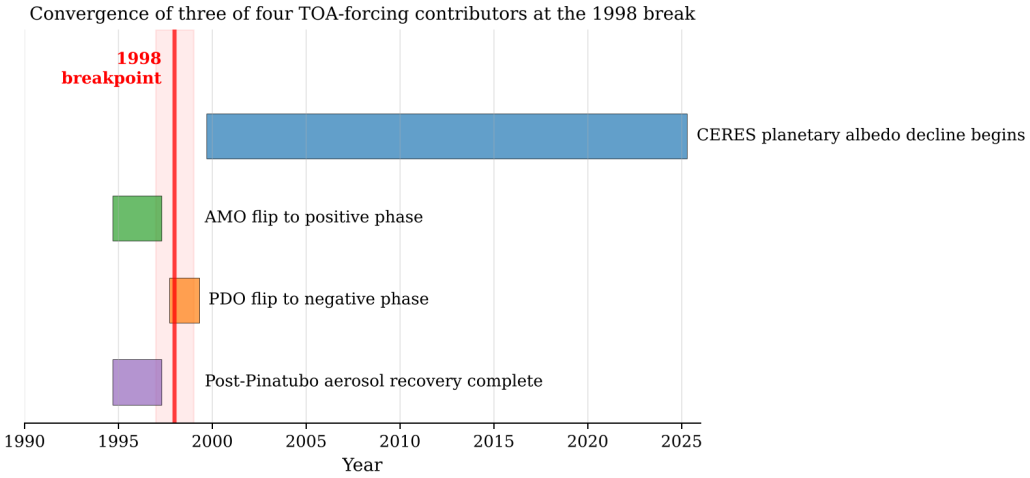


Figure 4: Three of the four TOA contributors identified in Cohler & Soon (2026b) [5] acquire step-change character within two years of the 1997–1998 Keeling breakpoint. The CERES-measured planetary albedo decline begins at 2000. The AMO flipped to positive phase 1995–1997. The PDO flipped to negative phase 1998–1999. The post-Pinatubo stratospheric aerosol rebound (Pinatubo eruption 1991) completed 1995–1997. The fourth contributor, TSI, is not sharply anchored at 1998 and is the smallest of the four by budget share.

The physical chain is kinetic. Ocean-surface pCO_2 responds to sea-surface temperature on the van 't Hoff Henry's-law timescale (months), and the enhanced outgassing flux carries the standard

δ^{13}C value near −8 ‰. Biosphere respiration responds to temperature on the Arrhenius Q_{10} law but is buffered by soil-carbon pool turnover on multi-decadal timescales, and the respired carbon it returns carries the much more negative C3 biomass value near −27 ‰. A simultaneous step in positive TOA forcing (albedo decline + AMO positive + PDO negative + post-Pinatubo recovery) produces an immediate ocean-outgassing response and a delayed biosphere-respiration response. The flux-weighted non-fossil $\delta^{13}C_N$ therefore shifts sharply toward the less-negative ocean value on the fast timescale, and b_K jumps accordingly. The biosphere offset that would partially drag b_K back toward the C3 value arrives on the slower timescale and has not yet closed the gap. The post-1998 window captures the system in the forward (ocean-dominated) phase of this kinetic separation. The one TOA contributor that does not sharply anchor at 1997–1998 is TSI: Solar Cycle 23 peaked near 2000 and Solar Cycle 24 (peak 2014) was anomalously weak, so the sustained TSI contribution since the breakpoint has been slightly negative rather than positive; TSI is also the smallest of the four in the budget (0.1–0.5 W m^{-2}).

The kinetic chain just stated is a consistency statement with the first-principles radiative and reservoir analysis of Cohler & Soon (2026b) [5], not an independent derivation from the isotopic series presented here. The isotopic series by themselves establish (i) that a significant structural break occurred at 1997–1998, (ii) that the shift is +0.98 ‰ in the direction of the less-negative ocean standard value, and (iii) under the mass partition equation with standard δ^{13}C values, that the shift corresponds to a 5-percentage-point rise in the ocean share of the non-fossil net flux. What they do not establish on their own is the causal attribution of that shift to any particular TOA contributor; that attribution rests on the radiative budget of Cohler & Soon (2026b) [5] and on the kinetic timescales summarized above.

5.3 Relation to the three prior isotopic falsifications

Four isotopic constraints on the fossil fraction f of the net atmospheric addition can now be stated as a single consistent ensemble: (i) stationarity of b_K at -13.2 ± 0.2 ‰ over 1978–2022 on the Scripps network and across the five-century sclerosponge proxy of Böhm et al. (2002) [2] [13]; (ii) algebraic bound $f \lesssim 0.013$ from the mass partition equation [4]; (iii) Hurst–Kolmogorov recent-window trend +2.94 ‰ per decade, $p < 10^{-4}$, positive sign opposite to the fossil prediction [6]; and (iv) the present extension to 2025 ($b_K = -13.42 \pm 0.07$ ‰ stationary on the independent Francey/Cape Grim record over 1020 years, with a significant 1997–1998 regime shift of +0.98 ‰ toward the ocean endmember during the highest-emissions decades). The four results share the same primary datasets and the same standard δ^{13}C values used throughout the IPCC carbon-cycle assessment literature [3]. Each is capable of standing alone. Each engages a different observable. Their mutual agreement is a statement about the internal consistency of the isotopic record under the non-fossil-dominated framework of Cohler & Soon (2026b) [5], and of their joint inconsistency with the fossil-dominance framework asserted by Canadell et al. (2021) [3] and reasserted in Engelbeen et al. (2024) [8]. Cohler & Soon (2026d) [7] falsifies the latter report on two independent grounds. The accounting identities it relies on carry no causal evidence, by pure mathematical argument. And the direct empirical measurements it adduces run in the opposite direction. The present result is one more line of evidence in that falsification.

6. Conclusion

The Keeling-plot intercept b_K of the net atmospheric CO_2 input, extended from the 44-year Scripps instrumental window of Koutsoyiannis (2024) [13] to the 1020-year Francey Law Dome + Cape Grim record from 1006 to 2025, is -13.42 ± 0.07 ‰ (1σ), statistically identical to the Koutsoyiannis value and 15 ‰ removed from the fossil value. Every CGAA-containing sub-window from 1750–2025 to 1850–2025 lies within ±0.4 ‰ of the Koutsoyiannis central value.

A Chow / Andrews sup-F scan of the modern segment locates a structural break at 1997–1998

(sup-$F \approx 52$). A HAC-robust Welch-style test of the intercept difference across that break gives $\Delta b_K = +0.98$ ‰ at $t = 2.66$, $p = 0.018$; the shift is significant under either OLS or HAC standard errors. The shift carries b_K to -12.79 ± 0.05 ‰, in the opposite direction from the fossil-dominance prediction and in the predicted direction of the ocean-dominated TOA-forcing mechanism of Cohler & Soon (2026b) [5]. Under the mass partition equation with $f \approx 0$ and standard δ^{13}C values the shift corresponds to a 5-percentage-point rise in the ocean share of the non-fossil net flux. Three of the four TOA contributors in that mechanism acquire step-change character within two years of the breakpoint; the fourth (TSI) is not sharply anchored at the break and is the smallest by budget share. The extended stationary intercept, the 1997–1998 breakpoint, and the direction of the post-break shift form a fourth complementary isotopic falsification of fossil-fuel dominance, consistent with the three previously established in Koutsoyiannis (2024) [13], Cohler & Soon (2026a) [4], and Cohler & Soon (2026c) [6].

Data Used

Francey et al. (1999) [9] Tables 1–3:
https://b.tellusjournals.se/articles/10.3402/tellusb.v51i2.16269

Cape Grim CGO5011 δ^{13}C (WDCGG):
https://gaw.kishou.go.jp/

Cape Grim CO_2 (CSIRO/BOM):
https://www.csiro.au/en/Research/OandA/Areas/Assessing-our-climate/Greenhouse-gas-data

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Conflict of Interest

The authors declare no conflict of interest.

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